

JEewan GUL

Full-Stack Developer (MERN) | AI/ML Project Experience

punjabijeewan@gmail.com | 0330-0829982 | Shikarpur, Sindh, Pakistan | [linkedin.com/in/jeewan-gul](https://www.linkedin.com/in/jeewan-gul) | github.com/JeewanGul | jeewangul.github.io

PROFESSIONAL SUMMARY

Full-stack developer (React, Node.js, Flask, MongoDB/SQL) and founder of a tutoring institution serving 25+ students. Delivered a paid commercial desktop application now running the daily operations of a local trading business, alongside a production web app in daily use and an NLP chatbot deployed on Streamlit Cloud, combining hands-on software development with real-world operations, two completed internships, and teaching experience.

CORE SKILLS & TOOLS

- **Languages:** C++, Java, Python, JavaScript (ES6+), SQL
- **Web / Full-Stack:** React.js, Node.js, Express.js, Flask, MongoDB, HTML5, CSS3, REST APIs, SQLAlchemy
- **Desktop & Deployment:** PyWebView, PyInstaller (Windows packaging), Chart.js, offline-first architecture, thermal & laser print integration
- **AI / ML:** Machine Learning, NLP (TF-IDF), scikit-learn, Logistic Regression, Random Forest, Pandas, NumPy, Streamlit
- **Databases:** MongoDB, MySQL, SQLite, Database Management Systems (DBMS)
- **Tools & Practices:** Git, GitHub, VS Code, Postman, Agile, Software Development Lifecycle (SDLC), API Integration (Twilio)
- **Core CS:** Data Structures & Algorithms (DSA), OOP, Complexity Analysis, Problem Solving
- **Soft Skills:** Teaching & Mentorship, Institution Management, Technical Communication, Cross-functional Collaboration

PROJECTS

Shop Management System - *Paid Client Project (Delivered & Deployed)* | Python (Flask), SQLite, PyWebView, Chart.js

- Delivered as a first paid client engagement for a flour mill and rice trading business, replacing a two-stage paper process in which every transaction was written into a rough ledger and then copied by hand into the final one, cutting daily bookkeeping time by an estimated 80%
- Built dual-unit billing (one product sold by bag or kilogram with correct stock conversion), a unified party ledger netting sales against purchases into one balance, credit due-date tracking, and profit reporting as revenue minus cost of goods sold minus expenses
- Implemented receipt and statement printing across 58mm/80mm thermal, A5 and A4 in English, Urdu and Roman Urdu, and installed and configured the thermal and laser printers on site
- Added bank and mobile-wallet tracking, expense and labour ledgers, daily cash reconciliation, scheduled backups and PBKDF2-hashed login; packaged with PyInstaller as a Windows executable running entirely offline, with no subscription or cloud dependency

Intelligent FAQ Chatbot - *AI Semester Project (Live Demo)* | Python, scikit-learn, NLP, Streamlit

- Built and deployed an NLP-based chatbot for Sukkur IBA University FAQs using TF-IDF vectorization and Logistic Regression for intent classification, with cosine-similarity fallback for out-of-scope queries
- Delivered a live accuracy/precision/recall/F1 performance dashboard as part of a 3-member CSC-350 AI team; authored the accompanying technical report

Student Performance Prediction System - *Live Demo* | Python, scikit-learn, Streamlit

- Trained and benchmarked Linear Regression against Random Forest Regressor models, achieving an R-squared score of 0.79 and reducing MAE from 1.43 to 1.22 versus baseline
- Deployed an interactive Streamlit application for real-time student performance prediction

JTC Student Management System - *Live Demo* | Python (Flask), SQLAlchemy, SQL, REST APIs

- Architected a multi-role (admin, student, parent) web application covering attendance, fee automation with PDF receipt generation, homework distribution with file attachments, and quiz/exam module with MCQ and True/False auto-grading

- Implemented Excel-based bulk question upload, automated result computation, and WhatsApp notification delivery via Twilio API, cutting manual administrative work for a live 25+ student user base
- Configured the app as an installable Progressive Web App (PWA) with a native install prompt, letting students and parents add it to their home screen for app-like access without a native codebase

EXPERIENCE

Software Developer & Founder | *Jeewan Tuition Classes, Shikarpur* | 2023 - Present

- Identified manual attendance, fee tracking, and homework distribution as key operational bottlenecks, then architected and built a full-stack management system (Flask, SQLAlchemy, SQL) that now runs all daily operations for 25+ students
- Automated fee collection and receipt generation with PDF output, eliminating manual paperwork and cutting monthly billing admin time by an estimated 70%
- Integrated Twilio API to deliver automated WhatsApp attendance and homework reminders to parents, reducing missed-assignment follow-ups and improving parent visibility into student progress
- Founded and scaled a tutoring institution from concept to a live operation serving Classes 1-12 across 8 subjects, personally designing curriculum, exam papers, and tiered assessments
- Redesigned programming instruction using structured, step-by-step practice modules for 25+ students, driving a 90%+ course completion rate

Data Analysis & Graphic Design Intern | *Pakistan Hindu Council* | Jul 2026 - Aug 2026 (Completed)

- Researched and compiled a database of 10 prospective sponsor companies, then conducted direct outreach calls to pitch sponsorship opportunities on behalf of the organization
- Redesigned the cover and expanded multiple sections of the PHC_Magazine_2025-2026 publication, including layout design and image integration
- Authored strategic research reports - an analysis of PHC's youth engagement contests proposing six new contest concepts, and a mapping of US-funded (USAID/Embassy) grant opportunities aligned to PHC's programmatic strengths
- Completed minority-literature book review assignments supporting the organization's cultural and community-awareness content

Communications Intern | *Press Network of Pakistan (PNP)* | Jul 2026 - Aug 2026 (Completed)

- Produced weekly public-awareness deliverables - research-backed presentations, close-ended survey questionnaires, and analytical write-ups - on assigned civic topics under weekly deadlines
- Wrote and independently submitted an opinion letter on the urban-rural education gap to eight national newspapers as a public-outreach writing task
- Practiced source verification by fact-checking a live online claim using AFP, AP, and Reuters fact-check methodologies
- Maintained structured weekly activity logs and professional correspondence with the supervising editor across the full six-week internship

EDUCATION

Bachelor of Science in Computer Science | *Sukkur IBA University* | CGPA: 3.41 / 4.0 | 2023 - Present

- Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming, Artificial Intelligence, Software Engineering, Theory of Computation

CERTIFICATIONS & ACTIVITIES

- **Award:** 2nd Place among 250+ participants, ICT Quiz Competition, SIBA Fest - PKR 10,000 cash prize
- **Certifications:** Google Agile Essentials Specialization (Google/Coursera, 2026); Google AI Essentials (Google/Coursera, 2026); Microsoft Student Ambassadors - Azure AI and Azure OpenAI Basics Workshop (Microsoft, 2026); Introduction to MongoDB (MongoDB, Inc., 2026); Programming in C++ (CODIO, 2024); Intro to Java & OOP (UPenn/Coursera, 2024); Google Prompting Essentials (Google, 2025); Intro to Machine Learning (Kaggle, 2025)
- **Activities:** Coding competitions, hackathons, volunteer teaching, sports organizing
- **Languages:** English (Professional), Urdu & Sindhi (Native)